

Graphs of Sine and Cosine (5.1)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $\frac{1}{2}$

_____ State the amplitude of $y = 6 \sin x$.

ANSWER: 4π

_____ State the phase shift of $y = \sin\left(x - \frac{\pi}{3}\right)$.

ANSWER: π

_____ State the midline of $y = \sin x + 4$.

ANSWER: -1

_____ State the period of $y = \sin(3x)$.

ANSWER: 6 # _____ State the period of $y = \sin(2x)$.	ANSWER: 3 # _____ State the minimum value of $y = 2 \sin x + 1$.
ANSWER: 5 # _____ State the period of $y = \cos\left(\frac{x}{2}\right)$.	ANSWER: $\frac{2\pi}{3}$ # _____ State the amplitude of $y = \frac{1}{2} \cos x$.
ANSWER: $y = 4$ # _____ State the amplitude of $y = -5 \cos x$.	ANSWER: $\frac{\pi}{3}$ right # _____ State the maximum value of $y = 2 \sin x + 1$.